

POSTER: Certified Runtime Switching for Adversarially Robust Control

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Abstract. Neural network policies are vulnerable to adversarial perturbations of their observations, and existing defenses such as adversarial retraining impose a permanent performance cost even during non-attacked operation. We propose a certified runtime switcher that selects between a high-performance policy and a robust backup at each timestep, using a binary classifier certified using randomized smoothing. To enable fast, real-time certification, we design a quadratic-activation switcher network whose output distribution allows exact analytical certification. On HalfCheetah, our switcher retains over 98% of clean performance while recovering over 82% of the reward lost to adversarial attacks.

1 Introduction

Neural network policies trained with reinforcement learning (RL) have enabled complex control in cyber-physical systems. However, they inherit a vulnerability: adversarial perturbations, small, bounded modifications to the observation which can drastically degrade policy performance [1]. Zhang et al. [4] formalized this with the state-adversarial MDP (SA-MDP) [5] and proposed training an approximately optimal adversary, showing that retraining against the opponent yields robust, retrained policies. However, these policies sacrifice performance under normal operation for the sake of robustness. This imposes a permanent performance cost even during clean operation, making it suboptimal when attacks are intermittent or come in bursts.

An alternative is runtime switching: use a high-performance RL policy when observations are trustworthy, and fall back to a robust backup only when an attack is detected. The key challenge is making this detection certified, as an empirical detector offers no guarantees against adaptive adversaries. Randomized smoothing [2] provides this guarantee. By classifying over Gaussian-perturbed copies of an input, smoothing can certify a robust radius within which the prediction is provably stable for the smoothed classifier.

We propose a certified runtime switcher for RL policies. A binary classifier is trained on clean and adversarially perturbed observations. Then, smoothing is used to generate a certified radius which is used in a dual threshold-based controller that transitions between the high-performance, possibly unsafe policy and

a safe backup. The dual threshold mechanism suppresses false alarms while supporting recovery from repeated and multi-burst attacks without permanent commitment to either policy. For efficient certification, we use a quadratic-activation network which enables exact analytical certification by the Gil-Pelaez inversion, replacing Monte Carlo sampling with a deterministic, fast computation, enabling its use in runtime operations. We evaluate our method on the CartPole and HalfCheetah environments. We demonstrate that our certified switcher retains near-optimal performance under clean conditions while recovering the majority of lost reward under different attack patterns.

2 Methodology

2.1 Threat Model

We consider an ℓ_2 -bounded observation-space adversary. At each attacked timestep t , the agent observes $o_t = s_t + \delta_t$, where s_t is the true state and δ_t is an adversarial perturbation such that $\|\delta_t\|_2 \leq \epsilon$. We consider three different types of bounded-budget attacks: single-burst, multi-burst, and arbitrary. In single-burst, the adversary may attack any subset of consecutive timesteps, while multi-burst attacks multiple subsets, with cooldown periods between them. In the arbitrary pattern, the adversary attacks a random selection of timesteps. For fair comparison, we ensure that each attack type attacks the same number of timesteps in one trajectory.

The agent has access to two policies: a high-performance policy π_{perf} and a robust backup policy π_{backup} . Our goal is to select between these policies at each timestep using a certified detection mechanism, achieving near- π_{perf} returns under clean conditions while limiting damage during adversarial episodes.

2.2 Certification via Smoothing

We train a binary classifier $f : \mathbb{R}^d \rightarrow \{0, 1\}$ to distinguish clean observations (0) from adversarially perturbed observations (1), using a dataset of trajectories collected from policy rollouts. Training uses Gaussian noise augmentation at scale σ , preparing the classifier for the smoothing procedure.

Randomized smoothing. Given a binary classifier f and an input x , we define a *smoothed classifier* $g(x) = \arg \max_c \Pr[f(x + \eta) = c]$ where $\eta \sim \mathcal{N}(0, \sigma^2 I)$. Cohen et al. [2] showed that if the top-class probability $p_A = \Pr[f(x + \eta) = c_A] \geq \frac{1}{2}$, then g evaluates provably constant within an ℓ_2 ball of radius:

$$R = \sigma \Phi^{-1}(p_A) \quad (1)$$

where Φ^{-1} is the inverse Gaussian CDF. In other words, no perturbation of size less than R can change the smoothed classifier’s output.

Since the classifier can be arbitrarily complex, the standard approach estimates p_A by drawing Monte Carlo samples and applying a statistical bound. This unfortunately limits the method’s use in real-time applications.

Exact certification We develop a method to avoid Monte Carlo sampling entirely by using a classifier architecture whose output distribution under Gaussian noise can be computed in closed form. We develop a classifier, QUADMLP, consisting of a single hidden layer with a quadratic activation $\phi(z) = z^2 + z$:

$$f(x) = W_2 \phi(W_1 x + b_1) + b_2 \quad (2)$$

When the input is perturbed by Gaussian noise $x + \eta$, each pre-activation $W_1(x + \eta) + b_1$ is itself Gaussian. After the quadratic activation, each hidden unit becomes a noncentral chi-squared random variable, and the logit output $f(x + \eta)$ is a sum of these variables. This distribution has a characteristic function that can be written analytically.

The Gil-Pelaez inversion theorem [3] provides a general formula for recovering the CDF of a real-valued random variable from its characteristic function $\varphi(t)$:

$$\Pr[Z \leq z] = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \operatorname{Im} \left[\frac{e^{-itz} \varphi(t)}{t} \right] dt \quad (3)$$

where Z is the logit difference under noise.

Since we have the characteristic function of the logit difference in closed form, we evaluate this single one-dimensional integral numerically to obtain the exact class probability p_A , and from it the certified radius R in Eq. 1. This replaces thousands of forward passes with a single deterministic computation, taking approximately 3 ms on CPU, making per-timestep certification feasible for Half Cheetah which has a control period of 8 ms.

Switcher Certification At each timestep, we certify the observation as safe if two conditions hold: the smoothed switcher predicts clean and the certified radius exceeds a threshold δ . The parameter δ controls the tradeoff between sensitivity and specificity: a larger δ demands a stronger certificate to trust the observation, increasing detection of attacks at the cost of more false alarms on clean inputs.

2.3 Dual-Threshold Switcher

Rather than switching policies on a single timestep’s certification, we want sustained evidence of an adversary or the lack of one before transitioning between controllers. Therefore, the switcher operates in two modes. In performance mode, an unsafe counter increments when the observation is not certified safe and decrements otherwise. When the counter reaches the environment hyper-parameter K_{enter} , the controller switches to backup mode. In backup mode, a safe counter increments on each certified-safe observation and resets to zero otherwise; when it reaches the other hyper-parameter K_{exit} , the controller returns to performance mode. This suppresses false alarms exponentially as triggering a false switch requires K_{enter} consecutive misclassifications. This naturally supports recovery from repeated attacks without permanently committing to one policy.

Environment	Controller	Clean	Single-burst	Multi-burst	Arbitrary
CartPole	Always Perf.	500 ± 0	107 ± 42	450 ± 98	500 ± 0
	Always Backup	500 ± 0	500 ± 0	500 ± 0	500 ± 0
	Cert. Switcher	500 ± 0	500 ± 0	500 ± 0	500 ± 0
Half Cheetah	Always Perf.	7245 ± 99	4082 ± 1290	3168 ± 1661	5929 ± 1210
	Always Backup	5652 ± 46	5641 ± 50	5649 ± 52	5632 ± 44
	Cert. Switcher	7132 ± 125	6687 ± 135	6598 ± 1196	5874 ± 1934

Table 1: Median cumulative reward $\pm$ standard deviation over 30 episodes.

3 Evaluation

We evaluate our certified switcher on two environments: CartPole and HalfCheetah. In both, we compare three controllers: **Always Perf.** (high-performance policy only), **Always Backup** (robust policy only), and our **Certified Switcher**.

3.1 Environment Setup

CartPole. A PPO (Proximal Policy Optimization) agent was trained, for the performance policy with Stable-Baselines3, achieving the goal reward of 500 in all clean episodes. The backup is a LQR controller, which can come at a more expensive cost. The adversary uses a PGD ℓ_2 attack with $\epsilon = 1.0$, applied at a burst of 200 consecutive timesteps. A hyper-parameter search found the environment parameters: $\delta = 0.25$, $K_{\text{enter}} = K_{\text{exit}} = 5$.

HalfCheetah. The performance policy is a pre-trained PPO agent and the backup is an ATLA-retrained policy (Alternating Training of Learned Adversaries) [4]. The adversary uses the optimal attack of Zhang et al. [4] with ℓ_2 bound $\epsilon = 0.5$. The environment parameters were chosen as: $\delta = 0.2$, $K_{\text{enter}} = K_{\text{exit}} = 5$.

3.2 Results

In CartPole, the **Always Perf.** controller collapses to a median reward of 107 under attack, while our switcher maintains 500 in both clean and attacked conditions while using the performance controller for $\sim 60\%$ of steps. Arbitrary attacks don’t affect the environment significantly since sustained attacks are needed to cause the pole to tip over.

In HalfCheetah, **Always Perf.** achieves the highest clean reward at 7245 but degrades to 4082 under single-burst and 3168 under multi-burst attacks. **Always backup** is invariant to attacks (5652) but permanently sacrifices $\sim 22\%$ of the clean reward. Our switcher retains 98.5% of the clean reward (7132) while recovering 82% of the reward lost by **Always Perf.** under single-burst and 84% under multi-burst attacks. Figure 1 shows a trajectory and how the certified radius alerts the switcher when to change policies. Notably, the switcher

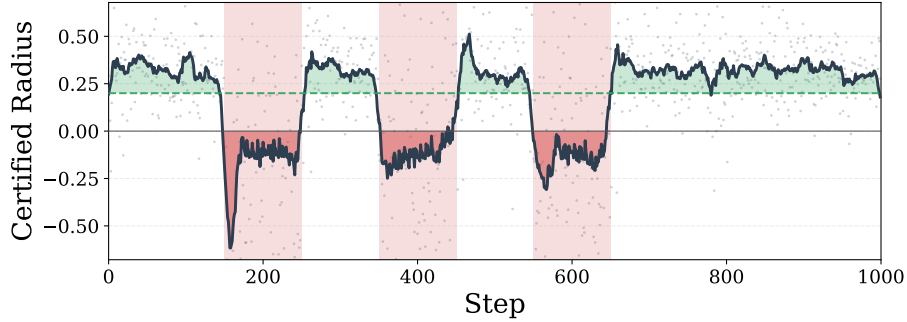


Fig. 1: 15-step rolling average of the certified radius on a Half Cheetah episode under a multi-burst attack. The radius stays above δ during clean intervals, keeping the performance controller active, and drops below zero during shaded attack bursts, triggering a switch to the backup controller.

exceeds **Always Backup** under all conditions, demonstrating that certified runtime switching outperforms permanently committing to the robust policy. Under arbitrary attack patterns, the switcher matches the other controller modes.

On HalfCheetah, which has an 8 ms control period, Monte Carlo randomized smoothing is too slow for real-time certification at practical sample counts. With 10,000 samples, its runtime is 13.41 ms, and with 100,000 samples, this increases to 104.51 ms, both exceeding the control budget. In contrast, our analytic certification method runs in 2.93 ms, remaining comfortably within the real-time constraint while providing an exact certificate with no sampling variance.

References

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